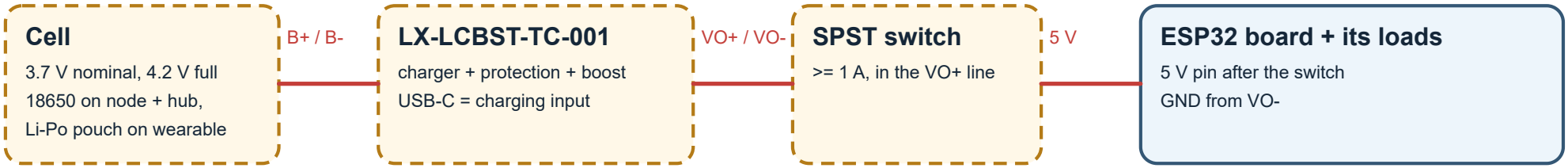
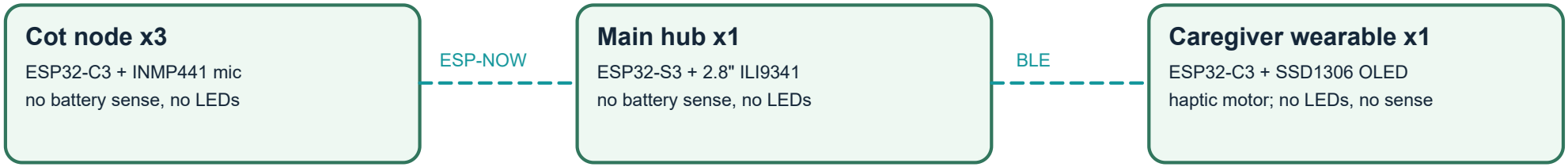
1 / CryNet system - power chain and links

Every unit has its own battery and its own ground. No wire ever joins two units.



One chain in all four units - only the loads differ:



Rules every unit follows:

1 - Switch OFF before charging. The cell still charges on B+/B-.

2 - Set the boost trimpot to 5.1 V with a meter before any board.

3 - Never power a board from USB and the boost at the same time.

Dashed teal = wireless link, no wire. Solid = a real conductor.

Proposed, bench-validation pending: the whole battery chain, the wearables and the LEDs. The node and hub signal wiring is the verified part. **No unit has a battery sense or a status LED (D-83)** - the node's divider and its GPIO10 blue link LED, the node's GPIO7 amber SOC LED, the hub's GPIO1 divider, the hub's GPIO2 BLE LED and the hub's GPIO17 SOC LED have all been removed. The hub's panel is the only display in the system, so nothing anywhere will warn the caregiver that a cell is nearly flat.

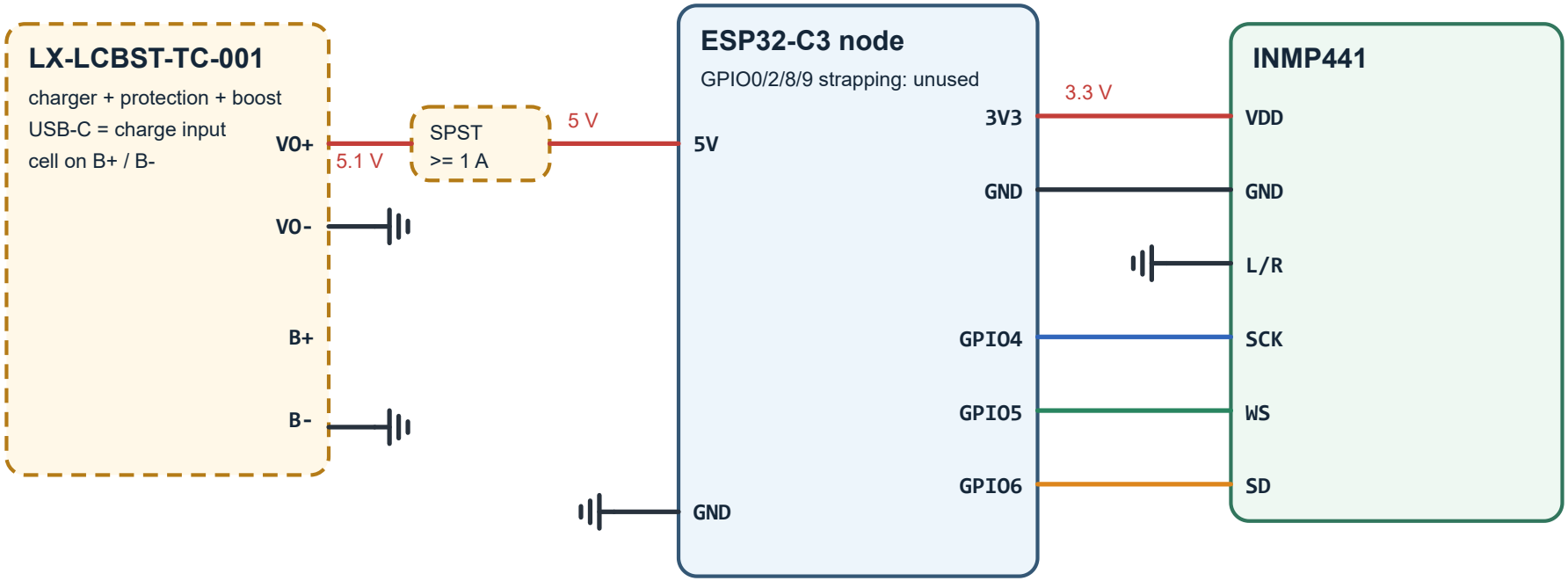
Cells (D-81): an 18650 (3.7 V, 4.2 V full, 3000 mAh) on each node and the hub, because a node sits on a cot frame within a baby's reach and needs a cell that cannot be punctured; a 1000 mAh Li-Po pouch on the wearable, because an 18650 is 18 x 65 mm and 48 g and cannot go on a wrist. **Never LiFePO4** - the module charges to 4.2 V and an LFP cell tops out at 3.65 V. The module's ~1 A charger must come down to ~400-500 mA for the wearable's pouch (1 C is too high for it).

Runtime (D-82): the chain boosts to 5.1 V and the board drops that to 3.3 V through a *linear* regulator, which passes current rather than power - so the cell supplies roughly 1.8x the 3.3 V load current. That projects to about six hours for the wearable on a 1000 mAh pouch. These are projections, not measurements. Feeding the 3V3 pin from a 3.3 V buck-boost would roughly double it - a change to the power chain, deferred to the bench.

No signal wire exists between the nodes and the hub - the link is wireless. Each unit is electrically independent; never join their batteries or grounds. Read with SYSTEM_WIRING.md sections 5 and 6.

2 / Cot node - complete wiring (one per node, x3)

ESP32-C3 SuperMini + INMP441 + LX-LCBST + switch - no battery sense, no status LEDs



No battery sense and no status LEDs (D-83).

B+ and B- carry the cell and nothing else: no divider,
no ADC read, no telemetry field, no battery text on the panel.

GPIO3, GPIO7 and GPIO10 are free.

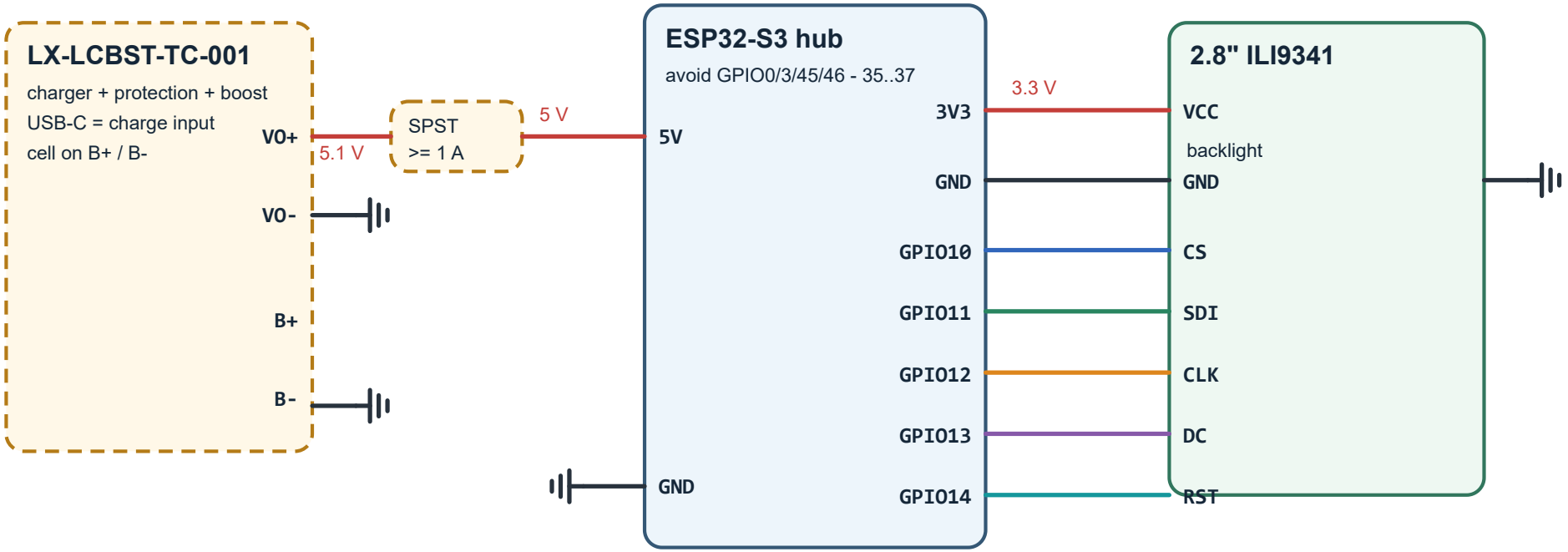
Verified on the live system: mic wiring (3V3 / GND / L/R→GND, GPIO4 SCK, GPIO5 WS, GPIO6 SD) and the node firmware.

No battery sense and no LEDs (D-83). The 220 kΩ / 100 kΩ divider, its 100 nF, and the planned GPIO10 blue link LED and GPIO7 red SOC LED are all withdrawn: the node does not read its cell, and the hub's panel already shows which cots are answering. This repository's node firmware did carry the divider read (PIN_BATTERY 3) until D-83 took it out; the LEDs were never in any firmware. **Do not fit a divider or an LED on this board.** GPIO1, GPIO3, GPIO7 and GPIO10 are free.

Every ⏚ symbol is the same net: the module's VO- (battery negative through the module's protection FET). Wires join only where a dot is drawn. Pin names, not header positions. Repeat identically for nodes 2 and 3.

3 / Main hub - complete wiring (x1)

ESP32-S3-DevKitC-1 N16R8 + 2.8" ILI9341 + LX-LCBST + switch - no battery sense, no status LEDs



No battery sense and no status LEDs (D-83).

GPIO1 (ADC), GPIO17 and GPIO2 are free.

Verified on the live system: TFT signals (CS 10, MOSI 11, SCK 12, DC 13, RST 14, BL 4, VCC 3V3, GND) and the hub firmware.

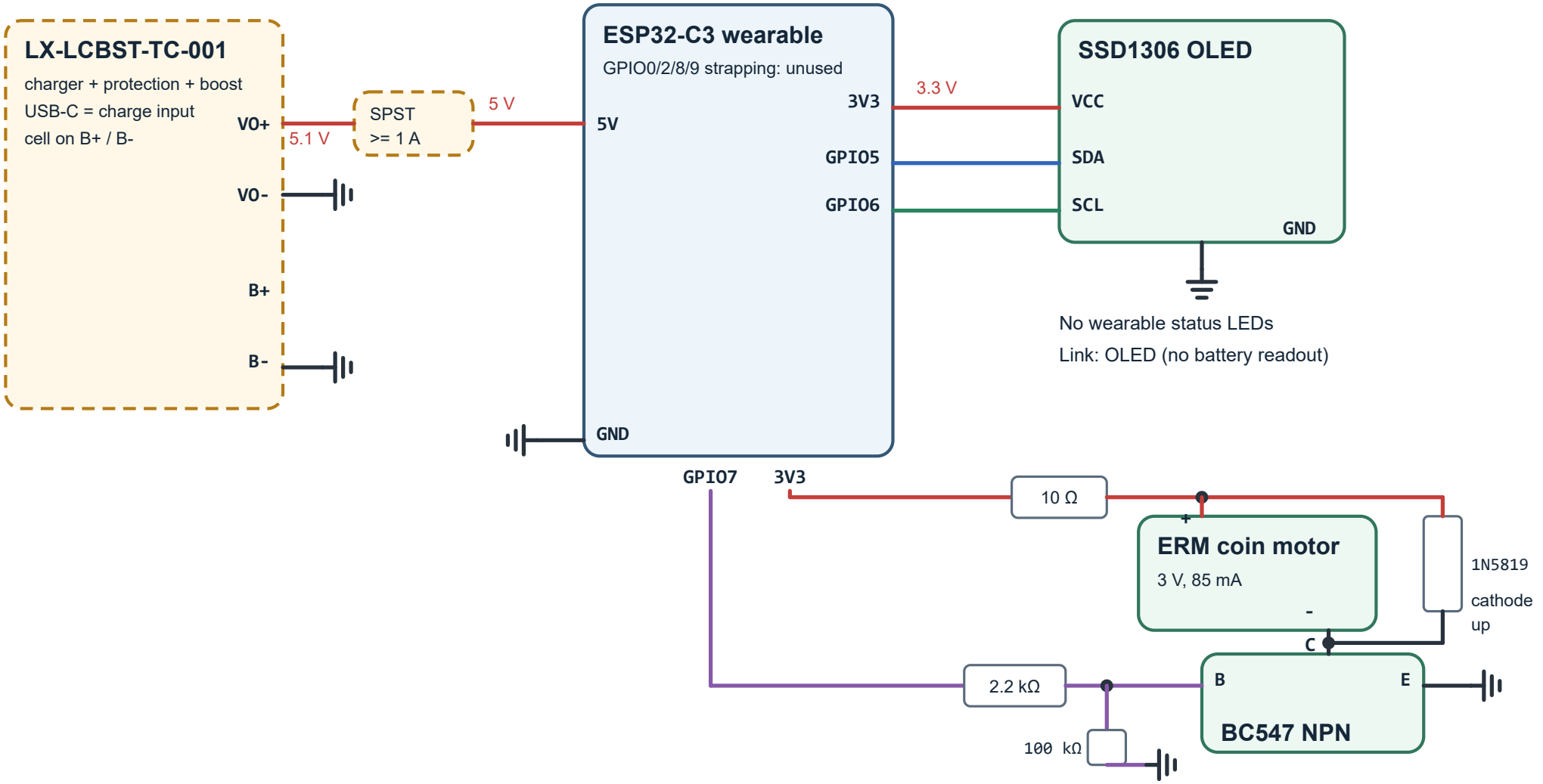
No battery sense and no LEDs (D-83). The GPIO1 divider (ADC1_CH0), the GPIO2 blue BLE-status LED and the GPIO17 red SOC LED have been removed before they were fitted. The hub does not measure its own cell and drives no indicator: the panel is the only display, and the wearable tells the caregiver what the hub is doing over BLE. **Do not fit a divider or an LED on this board.** GPIO1, GPIO2 and GPIO17 are free.

Proposed, needs bench validation: the battery chain. The hub's peak current with TFT + radios is unmeasured - confirm the LX-LCBST holds 5.1 V during bursts before trusting it here.

GPIO35-37 are the Octal PSRAM bus - never wire them. The TFT wiring is the recorded working one for the existing module. Every ⏚ symbol is the same net: the module's VO-.

4 / Caregiver wearable - complete wiring (x1)

ESP32-C3 SuperMini + SSD1306 OLED + LX-LCBST + switch + BC547 haptic drive (status on OLED)

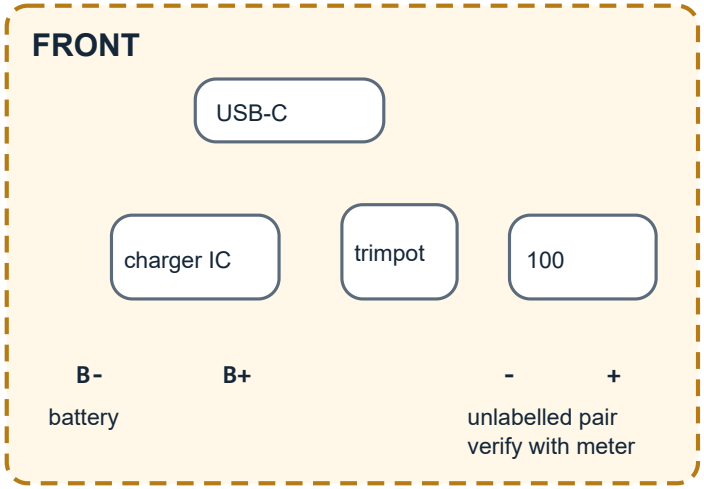


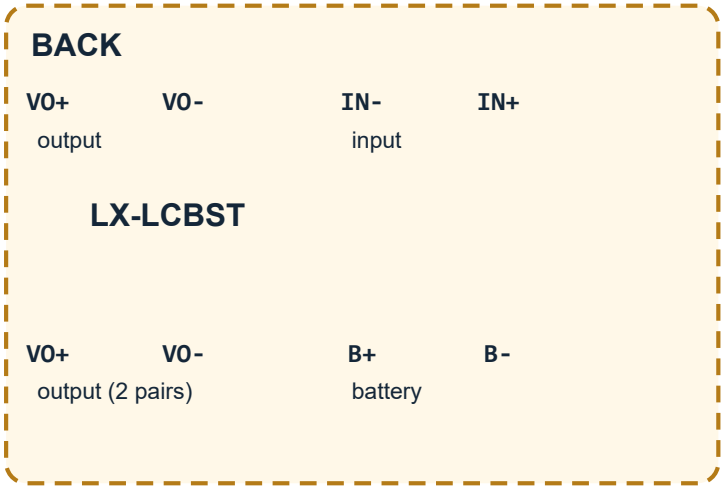
No battery sense and no LEDs (D-83). The 220 kΩ / 100 kΩ divider and its 100 nF are withdrawn, and the wearable has no indicator of any kind: the OLED carries LINK and SCAN, and says nothing about the cell. **Do not fit a divider or an LED on this board.** GPIO3 is free. Motor: the 10 Ω caps the stall current at 3.3 V ÷ (R + 10 Ω) - about 69 mA for a 37 Ω coin ERM, inside the BC547's 100 mA; a lower-impedance motor needs 22 Ω or a 2N2222A. GPIO7 drives the switch in 250 ms pulses. The OLED carries the state and the firmware already shows LINK or SCAN on it.

Every ⏚ symbol is the same net; wires join only where a dot is drawn. The BC547 emitter is the motor's ground, and the flyback sits across the motor with the cathode up.

5 / LX-LCBST-TC-001 - pad map and bench procedure

From the user's front-and-back photograph - Chinese silk text translated





Bench procedure, before any ESP32 is connected:

1 - Cell on B+/B-: turn the trimpot until VO+/VO- reads 5.1 V on a meter, no load.

2 - USB-C in: measure the charge current with a series meter - the cell must allow it.

3 - Confirm the charge ends at 4.2 V.

4 - Switch off: confirm the load is dead and measure the battery drain (boost quiescent unknown).

5 - Run one node and compare it with the USB baseline.

No T pad on this board - the listing's T-shorting notes belong to other variants. Never short anything.

Unknown until measured: the actual charge current (the program resistor is unreadable in the photo), the boost quiescent drain when off, the protection thresholds, and what the front - / + pair connects to. Buy ONE module for validation; only then buy the remaining three (plus the wearable's, pending the same check).